Class 3

P_1 \ P_2	C	D
C	(-1, -1)	(-5, 0)
D	(0, -5)	(-4, -4)

Defn A strategy s_i of player i is strictly dominated by another strategy t_i of the same player if $\forall\, s_{-i} \in S_{-i}$
$u_i(s_i, s_{-i}) < u_i(t_i, s_{-i})$

Assumption-1 a rational player will never play a strictly dominated strategies.

Assumption-2 All players in a game are rational.

$$S_{-i} = \underset{j \neq i}{\times} S_j$$

Example

P_1 \ P_2	L	M	R
T	1, 0	1, 2	0, 1
B	0, 3	0, 1	2, 0

↓

P_1 \ P_2	L	M
T	1, 0	1, 2
B	0, 3	0, 1

→

	L	M
T	1, 0	1, 2

↓

	M
T	1, 2

- Player 1 knows that Player 2 is rational.

Definition A fact is common knowledge among the players in a game if for any finite chain of players $i_1, i_2, \ldots, i_k$, the following holds: Player i_1 knows that Player i_2 knows that ... Player i_k know the fact.

Assumption-3 Rationality of the players is common knowledge.

Definition : A game is called complete information if the game is common knowledge.

For the first part, we will talk about complete information games.

Theorem The order of elimination of strictly dominated strategies do not matter, in the sense that the final game after removing all the strictly dominated strategies does not depend on the order of elimination.

- Write down the proof of the theorem.

P_1 \ P_2	L	C	R
T	1, 2	2, 3	0, 3
M	2, 2	2, 1	3, 2
B	2, 1	0, 0	1, 0

	L	R
T	1, 2	0, 3
B	2, 1	1, 0

Defn A strategy s_i of Player i is weakly dominated by another strategy t_i of the same player if

- $u_i(s_i, s_{-i}) \leq u_i(t_i, \ldots$

M	2,2	2,2
B	2,1	1,0

$\forall s_{-i} \in S_{-i}$

- $\exists\ t_{-i} \in S_{-i}$ s.t.
  $u_i(s_i, t_{-i})$
  $< u_i(t_i, t_{-i})$

	L	R
M	2,2	3,2

P_1 \ P_2	L	C	R
T	1,2	2,3	0,3
M	2,2	2,1	3,2
B	2,1	0,0	1,0

	L	C	R
M	2,2	2,1	3,2
B	2,1	0,0	1,0

	L
M	2,2
B	2,1

Order of elimination	Result
T, R, B, C	(M, L)
B, L, C, T	(M, R)
T, C, R	(M, L) or (B, L)